

CHANUTE, KS, USA

WMO: 724507

Lat: 37.670N

Lon: 95.484W

Elev: 300

StdP: 97.77

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13981

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.4	-10.8	-18.7	0.7	-11.7	-16.2	0.9	-9.5	12.0	7.1	11.3	7.1	3.6	350	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	36.7	24.0	34.8	24.4	33.2	24.2	26.2	32.6	25.6	32.1	25.0	31.3	5.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.3	20.0	30.1	23.6	19.1	29.3	23.0	18.4	28.5	83.0	32.8	80.2	32.2	77.4	31.3	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	10.3	8.9	DB	-18.0	38.3	3.5	2.4	-20.5	40.0	-22.6	41.3	-24.5	42.7	-27.1	44.4	(4)
(5)			WB	-18.3	27.2	3.3	0.7	-20.7	27.7	-22.7	28.1	-24.5	28.4	-27.0	28.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	0.8	3.1	8.6	14.1	19.1	24.1	26.7	26.0	21.6	14.8	8.2	2.3
	DBStd	10.31	6.14	6.22	6.00	5.01	4.21	3.14	2.96	3.14	4.17	5.06	5.51	5.70
	HDD10.0	972	291	207	97	19	1	0	0	0	0	15	96	246
	HDD18.3	2415	544	428	304	142	43	3	0	1	16	132	304	498
	CDD10.0	2494	4	12	55	143	283	422	517	497	348	163	43	6
	CDD18.3	895	0	0	3	16	68	174	258	239	113	22	1	0
	CDH23.3	9227	0	3	26	141	558	1729	2907	2577	1091	186	11	0
	CDH26.7	4063	0	0	3	24	143	683	1446	1282	439	42	1	0
Wind	WSAvg	4.2	4.3	4.6	5.0	5.2	4.3	4.1	3.5	3.4	3.6	4.1	4.4	4.3
Precipitation	PrecAvg	987	38	27	78	91	127	135	96	105	99	93	69	43
	PrecMax	1559	134	64	307	211	426	246	311	215	240	204	168	121
	PrecMin	541	2	2	7	13	46	14	1	22	1	3	2	2
	PrecStd	249	32	18	64	55	80	60	80	60	74	59	55	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	22.7	26.2	29.4	31.6	35.3	38.8	38.9	36.4	30.2	24.9	19.5
		MCWB	12.9	14.6	17.5	19.3	22.1	25.3	23.8	23.5	22.8	21.4	16.8	14.5
	2%	DB	16.1	18.9	23.2	26.8	29.8	33.3	36.8	36.8	33.0	27.7	21.8	16.3
		MCWB	10.1	13.2	15.7	18.2	21.8	24.5	24.2	23.7	23.6	19.3	15.1	11.9
	5%	DB	13.3	16.1	20.9	24.8	28.3	32.0	34.8	34.9	31.2	25.3	19.5	13.4
		MCWB	8.8	11.2	14.1	17.1	21.4	24.2	24.8	24.1	22.8	18.1	14.3	9.8
	10%	DB	10.1	13.0	18.6	22.8	26.6	30.6	33.2	33.0	29.1	23.1	17.0	11.0
		MCWB	6.0	8.8	13.3	16.4	20.5	23.4	24.7	23.8	21.4	17.8	12.5	7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	16.6	19.0	21.2	24.3	26.7	27.3	26.8	25.4	23.0	18.5	16.6
		MCDB	17.8	20.1	24.5	26.9	29.4	32.9	33.3	33.2	31.8	27.5	22.3	18.8
	2%	WB	11.9	14.2	17.2	19.9	23.3	25.7	26.3	25.8	24.6	21.7	16.7	13.1
		MCDB	15.0	18.1	21.3	24.4	28.0	31.7	32.7	32.7	30.7	25.4	20.3	15.3
	5%	WB	8.7	11.7	15.6	18.7	22.4	24.9	25.7	25.1	23.7	20.2	15.1	10.4
		MCDB	13.0	15.3	19.4	23.1	26.8	30.6	32.2	32.0	29.4	23.4	18.5	12.7
	10%	WB	6.4	8.7	13.7	17.3	21.3	24.1	25.2	24.4	22.7	18.4	13.2	7.8
		MCDB	10.5	12.9	17.7	21.9	25.5	29.2	31.7	30.6	27.8	22.2	16.4	10.6
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.2	12.0	12.0	11.1	10.8	11.3	11.8	12.2	12.1	11.8	10.6
		MCDBR	15.8	15.8	15.3	14.3	12.2	11.8	13.2	14.1	13.5	14.2	14.4	14.2
		MCWBR	10.4	10.4	9.0	8.1	6.2	5.3	4.2	4.4	5.6	7.3	8.8	10.0
	5% WB	MCDBR	13.8	14.1	12.2	12.2	10.5	10.8	11.2	11.6	11.5	10.8	12.2	11.9
		MCWBR	10.0	10.1	8.3	8.0	6.2	5.5	4.6	4.9	5.2	6.8	8.7	9.8
Clear-Sky Solar Irradiance	taub	0.296	0.307	0.343	0.380	0.417	0.423	0.435	0.439	0.384	0.344	0.308	0.288	
	taud	2.471	2.454	2.398	2.318	2.275	2.279	2.293	2.263	2.377	2.469	2.512	2.517	
	Ebn at Noon	894	928	920	899	868	859	845	834	864	872	873	882	
	Edh at Noon	81	94	110	126	134	133	131	132	111	92	78	73	
All-Sky Solar Radiation	RadAvg	2.39	3.08	4.03	5.10	5.72	6.57	6.54	5.85	4.88	3.58	2.58	2.04	
	RadStd	0.23	0.47	0.26	0.35	0.53	0.54	0.43	0.36	0.37	0.43	0.28	0.20	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page